

# Retrieval-Augmented Generation for Scientific Literature Assistants

## Abstract

This paper presents a retrieval-augmented generation (RAG) assistant that answers questions about scientific documents. Instead of relying only on what a large language model memorised during training, the system retrieves relevant passages from a PDF at query time and supplies them to the model as context. We describe the full pipeline: document loading, text splitting, embedding, vector storage, retrieval and answer generation, and we discuss how each design choice affects answer quality and latency.

## 1. Introduction

Research organisations produce and consume enormous volumes of papers, reports and manuals. Reading all of them is impossible, and keyword search often misses relevant passages that use different vocabulary. Large language models can summarise and explain text, but they may hallucinate facts that are not in the source. RAG reduces this risk by grounding the answer in retrieved evidence from the user's own documents.

## 2. Method

Documents are first loaded page by page. Because a full page is too long to embed precisely and may exceed the context window of the model, the text is split into overlapping chunks of about one thousand characters. Each chunk is converted into a dense vector with an embedding model, so that chunks with similar meaning are close together in vector space. The vectors are stored in a vector database. At question time the query is embedded with the same model and the nearest chunks are returned by similarity search. These chunks are inserted into a prompt together with the question, and the language model writes the final answer.

## 3. Results

On a small internal benchmark the RAG assistant answered factual questions about the loaded papers more accurately than the same model without retrieval, and it could point to the passages that supported each answer. Chunk size and overlap were the most important hyper-parameters: very small chunks lost context, while very large chunks diluted the similarity signal and increased prompt length.

## 4. Conclusion

We conclude that a modular RAG pipeline built from standard components is a practical way to give researchers fast, grounded access to a large collection of documents. Future work will explore hybrid retrieval, re-ranking, and citations displayed in the user interface.